

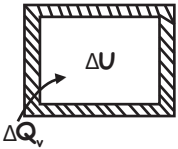
Internal Energy(U)

$$(1) \Delta U = \Delta Q_v = nC_v \Delta T = n \frac{f}{2} R \Delta T$$

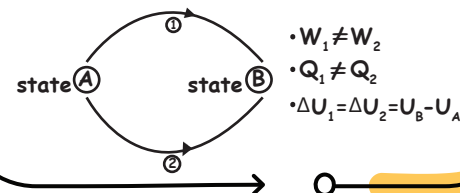
$$= \frac{nR \Delta T}{\gamma - 1} = \frac{\Delta(PV)}{\gamma - 1}$$

$$= \frac{P_f V_f - P_i V_i}{\gamma - 1}$$

(Internal Energy is only the function of temperature of the gas)



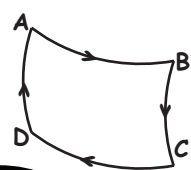
- First law of T.D $\Rightarrow Q = \Delta U + W$
- $Q, W \Rightarrow$ path functions
- $\Delta U \Rightarrow$ state function



Isothermal process $\Rightarrow \Delta T = 0 \Rightarrow \Delta U = 0$

Cyclic process

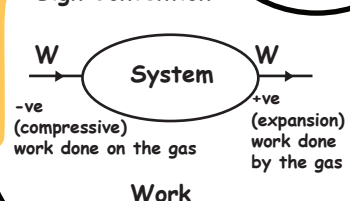
$A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$
 $\Delta U = U_A - U_A = 0$



WORK

Work done:
 path function
 $W = \int P dv$
 Unit: Joule(J)

Sign Convention



HEAT

Heat: path function
 $\Rightarrow$ unit: calorie/Joule

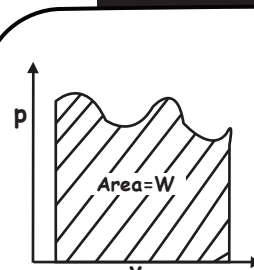
For any process,
 $Q = nC \Delta T$
 where, C = Specific heat capacity for the process

Adiabatic process $\Rightarrow \Delta Q = 0$ [No heat transfer]

At constant volume $\Rightarrow Q_v = \Delta U = nC_v \Delta T$

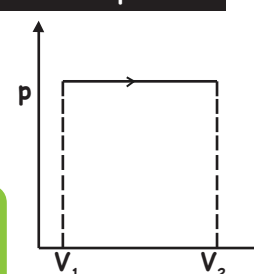
At constant pressure $\Rightarrow Q_p = \Delta U + W = nC_p \Delta T$

Work done from P-V Graph



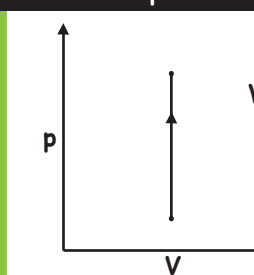
Area under P-V diagram gives work done by the gas

Isoobaric process

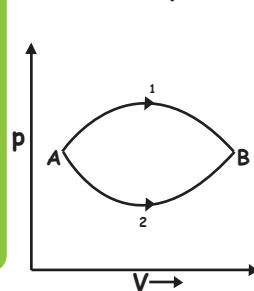


$$W = P(V_2 - V_1)$$

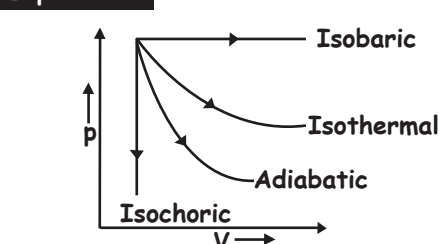
Isochoric process



$$W = P \Delta V = 0$$

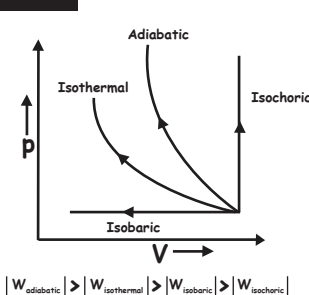


Expansion



$$W_{\text{isobaric}} > W_{\text{isothermal}} > W_{\text{adiabatic}} > W_{\text{isochoric}}$$

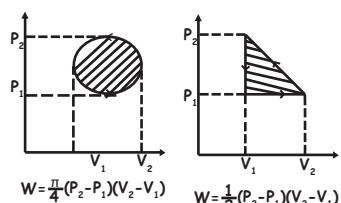
Compression



$$|W_{\text{adiabatic}}| > |W_{\text{isothermal}}| > |W_{\text{isochoric}}|$$

Cyclic process

- W = area inside the graph
- For clockwise process, $W = -ve$
- For anti-clockwise process, $W = +ve$



Thermodynamic processes

01 Isochoric process

- $Q = 0$ [no exchange of heat]

- Rapid or spontaneous process/insulated vessel

$$Q = \Delta U + W$$

$$Q = 0 \Rightarrow \Delta U = -W$$

Compression

$$W = -ve \quad \Delta U = +ve$$

$$\Delta U \uparrow \Rightarrow \text{Temperature} \uparrow$$

$$\Rightarrow \text{Pressure} \uparrow$$

Expansion

$$W = +ve \quad \Delta U = -ve$$

$$\Delta U \downarrow \Rightarrow \text{Temperature} \downarrow$$

$$\Rightarrow \text{Pressure} \downarrow$$

Equation of state

$PV^\gamma = \text{constant}$

$TV^{\gamma-1} = \text{constant}$

$$PT^{\frac{\gamma}{\gamma-1}} = \text{constant}$$

Work done by the gas

$$W = -\Delta U = nC_v(T_1 - T_2)$$

$$= n \frac{f}{2} R(T_1 - T_2)$$

$$W = \frac{nR}{\gamma-1}(T_1 - T_2)$$

$$W = \frac{P_i V_i - P_f V_f}{\gamma-1}$$

Slope of adiabatic process
 $= \gamma \times \text{slope of isothermal process}$
 specific heat of gas $\Rightarrow C = 0$

$$C = \frac{Q}{\Delta T} \rightarrow Q = 0$$

$$C = 0$$

Isothermal process

$$\Rightarrow \Delta T = 0 \Rightarrow \Delta U = 0$$

eg:- perfectly conducting
 slow process

$$Q = \Delta U + W$$

$$Q = W$$

equation of states $\Rightarrow PV = \text{Constant}$
 $P_1 V_1 = P_2 V_2$

$$\text{Workdone by the gas}$$

$$W = 2.303 nRT \log \left(\frac{V_2}{V_1} \right)$$

$$W = 2.303 nRT \log \left(\frac{P_1}{P_2} \right)$$

Slope:

Slope of adiabatic process
 $= \gamma \times \text{slope of isothermal process}$
 specific heat $C = \infty$

03 Isochoric process

$$\Delta P = 0$$

$$Q = \Delta U + W$$

equation of state $\Rightarrow V \propto T$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

Work done by the gas

$$W = P \Delta V = P(V_2 - V_1) = nR(T_2 - T_1)$$

specific heat

$$\Rightarrow C_p = \left(1 + \frac{f}{2} \right) R$$

$$= \frac{\gamma R}{\gamma - 1}$$

04 Isochoric process

$\Delta V = 0$ or $V = \text{constant}$

$$\text{equation of state} \Rightarrow P \propto T \Rightarrow \frac{P_1}{P_2} = \frac{T_1}{T_2}$$

Work done by the gas

$$\Delta V = 0 \Rightarrow W = 0$$

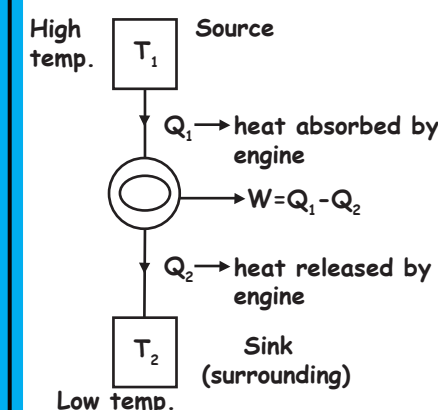
specific heat

$$\Rightarrow C_v = \frac{f}{2} R$$

$$= \frac{R}{\gamma - 1}$$

Heat Engine

'Device that converts heat into work'



$$\text{efficiency}(\eta) = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

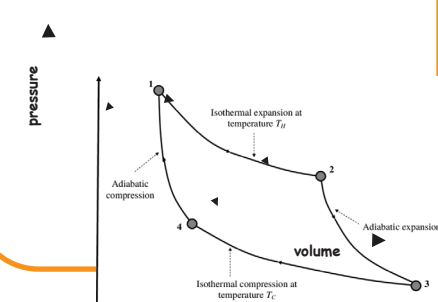
$$\eta_{\text{Carnot}} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

$\eta_{\text{max}} \Rightarrow$ When $Q_2 = 0$ or $T_2 = 0K$ (not possible)

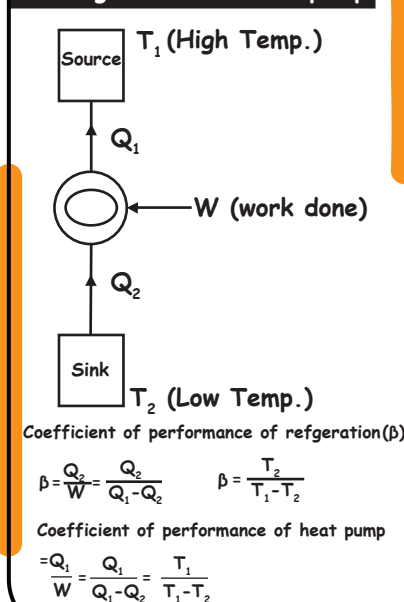
Carnot Engine

$\rightarrow$ Ideal engine

$$\eta = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$



Refrigerator and heat pump



Coefficient of performance of refrigeration(β)

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} \quad \beta = \frac{T_2}{T_1 - T_2}$$

Coefficient of performance of heat pump

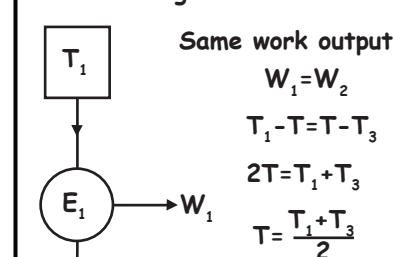
$$= \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} = \frac{T_1}{T_1 - T_2}$$

Relationship between

$$\beta = \frac{1 - \eta}{\eta}$$

$$(COP)_{\text{heat pump}} = 1 + (COP)_{\text{refrigerator}}$$

Cascaded engine



Same work output

$$W_1 = W_2$$

$$T_1 - T_2 = T_2 - T_3$$

$$2T_2 = T_1 + T_3$$

$$T_2 = \frac{T_1 + T_3}{2}$$

Same efficiency

$$1 - \frac{T_2}{T_1} = 1 - \frac{T_3}{T_2}$$

$$T_2^2 = T_1 T_3$$

$$T_2 = \sqrt{T_1 T_3}$$



THERMODYNAMICS